Intervening-halo DM for the 55 FRBs with DM_{host}

(a) Where the observed DM comes from

observed DM; DM_{MW}; $\langle DM_{cosmic} \rangle$; foreground halos; host (after foreground); host term < 0

DM (pc cm^{-3}, observer frame)

(b) DM from intervening galaxy halos

spectroscopic redshifts only (lower bound)
random sightlines 0.75° from the host (mean of about 2)
foreground halos (photo-z + spec-z)

median over FRBs: own sightline 46, mean of its random sightlines 38 pc cm^{-3}; > 100 pc cm^{-3}: 13 of 49 FRBs
in the host rest frame, $DM_{fg}(1+z)$: median 58, > 100 for 15 of 49

DM_{fg} (pc cm^{-3}, observer frame)

(c) Host DM before and after removing the foreground halos

DM_{host} (standard)
DM_{host}, foreground halos removed
predicted DM of the host's own halo

median foreground share of DM_{host}: 24%; negative after removal: 6 of 49

DM_{host} (pc cm^{-3}, rest frame)

(d) Halo crossings; expected number of sightlines with at least one: 35.9 of 49

P(at least one foreground halo crossed)
numbers: likely crossings with $p_{fg} \geq 0.84$

probability

20250316A (0.01) 20171020A (0.01) 20220319D (0.01) 20240210A (0.02) 20231120A (0.04) 20220207C (0.04) 20211127I (0.05) 20240312D (0.05) 20211212A (0.07) 20240203D (0.07) 20220509G (0.09) 20230124A (0.09) 20240318A (0.11) 20220914A (0.11) 20190608B (0.12) 20240213A (0.12) 20230628A (0.13) 20210807D (0.13) 20210410D (0.14) 20231226A (0.16) 20230526A (0.16) 20220920A (0.16) 20220725A (0.19) 20221106A (0.20) 20240215A (0.21) 20191001A (0.23) 20190714A (0.24) 20240208A (0.24) 20220825A (0.24) 20240304A (0.24) 20220307B (0.25) 20231123B (0.26) 20230307A (0.27) 20210320C (0.28) 20221012A (0.28) 20240229A (0.29) 20190102C (0.29) 20180924B (0.32) 20240525A (0.33) 20230626A (0.33) 20231220A (0.34) 20211203C (0.34) 20230902A (0.36) 20200906A (0.37) 20240224A (0.37) 20240119A (0.38) 20220501C (0.38) 20230613A (0.39) 20230907D (0.46) 20181112A (0.48) 20231020B (0.48) 20220310F (0.48) 20221219A (0.55) 20220418A (0.62) 20190523A (0.66)

FRB (host redshift), ordered by redshift; grey columns: outside the Legacy Survey footprint, no foreground estimate